Three-Rate Gated Memory for Test-Time Deep MIMO Receivers: Physics-Based Gates and Clipped Consolidation Against Pilot Poisoning

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***Abstract*—A neural receiver that learns at test time trains on pilots an adversary can write to. This paper argues that it should consolidate into persistent memory only what a label-free physical check verifies against that memory. Citadel-Rx applies this to DeepSIC with a fast level fitted per block and two slow levels updated by a gated, clipped average of the faster states; the clip bounds how far any attacker moves the slowest level per consolidation. With a single memory, a cyclic redundancy check (CRC) test with rollback fails, as do the gates of Citadel-Rx, because their only reference, a linear minimum mean-square error (LMMSE) detector on the poisoned pilots, degrades with the attack. On ten held-out seeds, white-box poisoning at 0 dB signal-to-interference ratio raises the symbol error rate of Citadel-Rx by $+0.0188$ over its clean run, against $+0.768$ for tuned online DeepSIC; the gates reject 99% of the poisoned blocks at a 0.1% false-rejection rate. Attacks adapted to the gates or their feedback leave the excess unchanged within its intervals, and the ordering holds with a second backbone and with 8×32 arrays. Attacks on the references or during a channel change can raise the excess, and the receiver fails at 16-QAM and in a coded 5G link over 3GPP urban-micro channels, where its network passes far fewer CRCs than LMMSE on clean blocks.**

***Index Terms*—Adversarial pilot poisoning, continual learning, deep receivers, MIMO detection, physical-layer security, physics-informed gating, test-time learning.**

I. Introduction

DEEP receivers replace or augment model-based receiver blocks with trained networks [1], [2]; since a network trained in simulation meets a different channel in the field, the strongest adapt at test time: online DeepSIC [3] re-trains its per-user networks on every block, and Meta-DeepSIC [4] meta-learns an initialization from past blocks. In [4] the training data are decoded, re-encoded codewords and no new pilots are sent. This paper studies the pilot-trained form attacked in [5], where a pilot perturbation crafted on a surrogate transfers to both receivers and to two other detectors. In the runs of Section VI-A, white-box poisoning at a signal-to-interference ratio (SIR) of 0 dB raises the symbol error rate (SER) of online DeepSIC by $+0.768$, and Meta-DeepSIC, which keeps the poisoned pilots in its support set, still carries an excess SER of $+0.0882$ after the attack.

The poisoning problem addressed here is the following. U single-antenna users send QPSK symbols to an N-antenna base station in blocks with N_p pilots per user and data framed by a cyclic redundancy check (CRC). The receiver adapts on the pilots of every block, and an adversary perturbs the received pilots of a window of blocks within a Frobenius-norm budget. The receiver must keep its SER close to that of the same run without attack, during and after the window, track legitimate channel change and use no labels beyond the pilots and the CRC.

Four properties of this problem shape the design. First, the pilots are the labels, but the physical consistency of an update can be checked through the fit of the decisions to a linear model, the CRC and the channel innovation. Second, the harm lies less in the attacked block than in what the receiver keeps. Third, a check against a reference that the attacker also degrades fails: a CRC test with rollback, judged against the linear minimum mean-square error (LMMSE) detector on the same pilots, accepts the first poisoned update as its rollback state. Fourth, only adapted attacks [6]–[8], matched clean runs and held-out seeds can show the value of a defense.

Pilot-spoofing detectors [9], [10] flag an established threat [11], [12] but protect no learning receiver. Code-verified decision-directed training [13], [14] verifies labels, not what enters a persistent memory, and drift detection [15] addresses channel variation, not an adversary. In vision, test-time adaptation (TTA) filters, anchors or restores what enters the weights [16]–[19], anchors fine-tuning to a prior, as transfer and continual learning do [20], [21], or averages a teacher, as semi-supervised learning does [22]. TTA has its own poisoning problem [23], [24], smaller when the attacker sees neither the online weights nor other users' samples [25], and its own robust-statistics defenses [26]; none uses a physical check. Continual learning under poisoning [27] treats regularization-based learners over a sequence of tasks, whereas the learner here sees one correlated block at a time. Nested learning [28] and Titans [29] learn memory at test time but have no test that can refuse a write. Table I summarizes the gap: to the authors' knowledge, no published deep receiver combines multi-rate memory, physics-gated consolidation and a retention bound.

The principle tested here is that a test-time receiver resists pilot poisoning when it consolidates into persistent memory only what an independent, label-free physical check verifies against the protected memory. It is a design condition supported by an ablation, not a guarantee.

The contributions of this paper are as follows.

1) **A receiver that consolidates only what physics verifies** (Section III): gated, clipped consolidation into two slow levels, an anchor fallback and CRC selection. In the

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TABLE I
POSITIONING BY METHOD CLASS (TTA: TEST-TIME ADAPTATION; SURFACE: WHAT THE ADVERSARY WRITES TO; †PILOT-TRAINED FORM ATTACKED IN [5]; –: NONE)

Class	Adapts	Memory	Surface / defense
Spoofing detection [9], [10]	no	–	pilots / detection only
Online DeepSIC† [3]	per block	last block	pilots / –
Meta-DeepSIC† [4]	per block	pilot buffer	pilots, persistent / –
Code-verified training [13], [14]	decoded data	last block	labels / code check, no persistent gate
CRC-Rollback	per block	last accepted	pilots / CRC gate, rollback
Filtering TTA [16]–[19]	per batch	weights, anchor	batch / entropy filter, restore
Robust TTA [23], [26]	per batch	batch statistics	batch / robust statistics
Anchored tuning [20]–[22]	per step	prior, average	data / pull to a prior or teacher
Nested memory [28], [29]	per token	one or several rates	input / –
Citadel-Rx	per block	three rates, gated	pilots / physics gates, protected fallback, bounded retention

ablation, a slow level, the fallback and a decision-level gate are necessary (excess SER $+0.316$, $+0.0859$ and $+0.174$ without each, against $+0.0199$).

2) **Retention bounds for the algorithm as implemented** (Section IV): a design bound on the slowest level with no measured quantity under any attacker and a structural bound for unequal verdicts, after correcting the consolidation rule of an earlier version of this receiver (the first version hereafter), which broke the convexity they need.
3) **Relative physics gates with a CRC design rule** (Sections III-C and III-E): three label-free ratio tests and a CRC threshold set by a target excess symbol-error probability.
4) **An evaluation with tuned baselines and adapted attacks** (Sections V and VI): anchored, decision-directed, EATA- and CoTTA-style baselines, adapted and physical attackers, a New Radio (NR) low-density parity-check (LDPC) coded link over 3GPP drops, a second backbone and 8×32 arrays; settings fixed on calibration seeds (two exceptions, Section V), held-out seeds, matched clean runs, bootstrap intervals and seed-level tests.

The evaluation also reports, with their mechanisms, where the method does not help: reference degradation through the data, denial of adaptation during a legitimate change, and 16-QAM and the coded scene, where the network is weaker than LMMSE on clean blocks. Sections II–IV give the model, the receiver and its bounds, Sections V–VII the evaluation, and Section VIII concludes.

II. SYSTEM MODEL AND THREAT MODEL

A. Uplink and Adaptive Receiver

U single-antenna users send unit-energy QPSK symbols from $\mathcal{X}$ to N base-station antennas. In block $b = 0, 1, \ldots$ the channel $\boldsymbol{H}_b \in \mathbb{C}^{N\times U}$ is constant, and slot t receives

$$\boldsymbol{y}_t = \boldsymbol{H}_b \boldsymbol{s}_t + \boldsymbol{w}_t, \qquad \boldsymbol{w}_t \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N), \tag{1}$$

with $\sigma^2 = \frac{1}{2}\,10^{-\mathrm{SNR}/10}$ for the signal-to-noise ratio (SNR) in dB. The N_p pilot slots form $\boldsymbol{Y}_b \in \mathbb{C}^{N_p\times N}$ with known symbols $\boldsymbol{S}_b$, and the $N_d = JL$ data slots are framed per user in J CRC sub-blocks of L symbols. The main channel is the spatial exponential decay (SED) model [3] with cosine fading,

$$[\boldsymbol{H}_b]_{\nu u} = c_{u,b}\, e^{-|\nu-u|}, \qquad c_{u,b} = 0.8 + 0.2\cos(2\pi b/P_u), \tag{2}$$

for antenna ν, user u and period P_u (stretched to $e^{-|(\nu-1)U/N-u+1|}$ for the larger arrays). DeepSIC [3] runs Q soft-interference-cancellation iterations, in which the one-hidden-layer multilayer perceptron (MLP) $f_{\theta_{u,q}}$ of user u maps $\boldsymbol{y}_t$ and the other users' posteriors to

$$\boldsymbol{\psi}_{u,t}^{(q)} = \mathrm{softmax}\, f_{\theta_{u,q}}\big(\boldsymbol{y}_t, \{\boldsymbol{\psi}_{u',t}^{(q-1)}\}_{u'\neq u}\big), \qquad q = 1, \ldots, Q, \tag{3}$$

from uniform $\boldsymbol{\psi}^{(0)}$; $\hat{s}_{u,t}$ maximizes $\psi_{u,t}^{(Q)}(s)$, and θ collects all weights. The stage and detector losses on pilots $\boldsymbol{Y}$ are

$$\begin{aligned} \ell^{(q)}(\theta; \boldsymbol{Y}) &= -\sum_{u=1}^{U} \frac{1}{N_p} \sum_{t=1}^{N_p} \log \psi_{u,t}^{(q)}(s_{u,t}), \\ \ell^{\mathrm{det}}(\theta; \boldsymbol{Y}) &= -\frac{1}{N_p U} \sum_{t=1}^{N_p} \sum_{u=1}^{U} \log \psi_{u,t}^{(Q)}(s_{u,t}), \end{aligned} \tag{4}$$

where $\ell^{(1)}$ feeds the other users' one-hot pilot symbols and $\ell^{(q)}$, $q > 1$, the posteriors of (3). Adapting stage q on $\ell^{(q)}$, in turn, defines the map $\theta_b = \Phi(\theta_{b-1}; \tilde{\boldsymbol{Y}}_b)$. Online DeepSIC starts from θ_{b-1}; Meta-DeepSIC [4] starts from a model-agnostic meta-learning (MAML) [30] initialization learned on the last 20 pilot blocks, refreshed every five blocks.

B. Threat Model

The adversary perturbs the pilots of an attacked block within a budget set by the SIR in dB,

$$\tilde{\boldsymbol{Y}}_b = \boldsymbol{Y}_b + \boldsymbol{E}_b, \qquad \|\boldsymbol{E}_b\|_F \le \epsilon_b = \|\boldsymbol{R}_b\|_F\, 10^{-\mathrm{SIR}/20}, \tag{5}$$

with $\boldsymbol{R}_b = \boldsymbol{Y}_b$ (noise included) or $\boldsymbol{S}_b \boldsymbol{H}_b^{\top}$ (signal only). At SIR 0 dB the attacker's received pilot energy equals that of all users (plus the noise when $\boldsymbol{R}_b = \boldsymbol{Y}_b$). The attacker is synchronized and knows the standardized pilots; white-box attackers know more than a transmitter can observe (Table II), which bounds from above what a causal attacker can know. The availability attacker A1 [5] solves

$$\begin{aligned} &\boldsymbol{E}_b^{\star} \in \arg\max_{\|\boldsymbol{E}\|_F \le \epsilon_b} \mathcal{O}_1(\boldsymbol{E}), \quad \mathcal{O}_1(\boldsymbol{E}) = U\, \ell^{\mathrm{det}}\big(\tilde{\theta}(\boldsymbol{E}); \boldsymbol{Y}_b\big), \\ &\tilde{\theta}_q \leftarrow \tilde{\theta}_q - \mu\, \nabla_{\theta_q} \ell^{(q)}\big(\tilde{\theta}; \boldsymbol{Y}_b + \boldsymbol{E}\big), \end{aligned} \tag{6}$$

where the surrogate $\tilde{\theta}(\boldsymbol{E})$ starts from the victim's weights ($\theta_{b-1}^{(1)}$ for Citadel-Rx) and applies the second line $T_{\mathrm{a}} = 2$ times per stage with $\mu = 5 \cdot 10^{-3}$; Adam, the delta rule and the proximal pull of Section III-A are not unrolled. Projected gradient ascent (PGD) [31] takes 20 steps

$$\boldsymbol{E} \leftarrow \Pi_{\epsilon_b}\Big(\boldsymbol{E} + \frac{\epsilon_b}{4} \frac{\nabla \mathcal{O}(\boldsymbol{E})}{\|\nabla \mathcal{O}(\boldsymbol{E})\|_F}\Big), \quad \Pi_r(\boldsymbol{v}) = \boldsymbol{v} \min\Big\{1, \frac{r}{\|\boldsymbol{v}\|}\Big\}, \tag{7}$$

from $\boldsymbol{0}$ and keeps the best iterate; restarts begin at random points of norm ϵ_b, and expectation over transformation (EOT) [32] averages over pilot noise. The soft-gate objective is

$$\begin{aligned} &\mathcal{O}_{\mathrm{sg}}(\boldsymbol{E}) = \ell^{\mathrm{d}}\big(\tilde{\theta}(\boldsymbol{E})\big)\, \tilde{\gamma}(\boldsymbol{E}) + \tilde{\gamma}(\boldsymbol{E}) - 1, \quad \tilde{\gamma} = \min\{\tilde{v}_1, \tilde{v}_2\}, \\ &\tilde{F} = \frac{1}{|\mathcal{J}_b|\, U} \sum_{j\in\mathcal{J}_b} \sum_{u=1}^{U} \prod_{t\in j} \psi_{u,t}^{(Q)}(s_{u,t}), \end{aligned} \tag{8}$$

TABLE II
ATTACKERS (–: NONE). WHITE-BOX: WEIGHTS, ADAPTATION MAP, GATE STATISTICS, RECEIVED PILOTS, NOISE POWER; + DATA: ALSO THE DATA AND ITS SYMBOLS; BOTH EXCEED WHAT A CAUSAL ATTACKER CAN KNOW

Attacker	Perturbation and objective	Knows
A1 [5]	PGD (7) on the post-adaptation loss (6)	white-box
A1, adapted	unroll 10 or 50; PGD 100, 5 restarts; EOT [32] soft gates (8), alone or combined	white-box + data
A2	A1 with the loss of user 1 minus that of the others	white-box
A4	share α on the pilots, $1-\alpha$ on the data (9); soft gates	+ data
A3*, slow	5, 20 or 50% of the pilot norm over 300 or 900 blocks; soft gates	+ data
Denial	spoof $\boldsymbol{S}_b\boldsymbol{Z}^\top$ of the channel estimate, $\boldsymbol{Z}$ random and fixed per run, scaled to ϵ_b, during a redraw	pilots
Physical	(10), $M=1,2,4$; $\boldsymbol{G}$ known or with error variance 0.1, 0.01	white-box
3GPP scene	physical, $M=1$, at 35, 75 or 150 m, a user's maximum power	white-box
Gray-box	A1 through a Citadel-Rx surrogate trained on observed pilots	settings
Transfer	A1 through an online DeepSIC surrogate	pilots
Query	search step on the NACK fraction [33], [34], one query per block, not converged	ACK/NACK
Jamming	Gaussian noise on the data at the same SIR (control)	–

with ℓ^{d} the cross-entropy on the true data of 64 random sub-blocks $\mathcal{J}_b$ (all for A4), $\tilde{F}$ a soft pass fraction, and $\tilde{v}_1$, $\tilde{v}_2$ soft verdicts of G1, against LMMSE only on posterior-mean symbols, and of G2 with $\tilde{F}$ for F_{nn}, from (20)–(24) with untightened thresholds and references from the unattacked block; G3 is not modeled. The soft verdicts, also used by A4 and A3*, replace the hard gates on the forward and backward pass. A4, with 6 PGD steps, splits the budget,

$$\|\boldsymbol{E}_b^{\mathrm{p}}\|_F \le \sqrt{\alpha}\,\epsilon_b, \qquad \|\boldsymbol{E}_b^{\mathrm{d}}\|_F \le \sqrt{1-\alpha}\,\epsilon_b^{\mathrm{d}}, \tag{9}$$

with ϵ_b^{d} from (5) on the data, at about $\alpha + (1-\alpha)N_d/N_p$ times the energy of A1 ($N_d/N_p \approx 250$). The physical attacker with M antennas and channel $\boldsymbol{G}$, i.i.d. $\mathcal{N}(0,1/N)$, designs for an estimate $\hat{\boldsymbol{G}}$ with relative error variance 0.1 or 0.01 (or $\hat{\boldsymbol{G}} = \boldsymbol{G}$), so that the designed and received perturbations are

$$\boldsymbol{E}_b = \boldsymbol{X}_b\hat{\boldsymbol{G}}^\top, \qquad \boldsymbol{X}_b\boldsymbol{G}^\top = \boldsymbol{E}_b\big(\boldsymbol{G}\hat{\boldsymbol{G}}^{+}\big)^\top, \tag{10}$$

with $\hat{\boldsymbol{G}}^{+}$ the pseudo-inverse; each PGD step is projected by $\boldsymbol{E} \mapsto \boldsymbol{E}(\hat{\boldsymbol{G}}\hat{\boldsymbol{G}}^{+})^\top$. Table II lists every attacker.

C. Metrics

The SER counts the data slots in which any user is wrong [3]–[5]. For S seeds with attack windows $\mathcal{W}_h$, and primes for the clean twin (same configuration, seed, channel draws and machine, attack disabled; for A4, a run on another machine that matches the attacked run block for block before the window), the SER, the excess SER, and the rejection and false-rejection rates are

$$\begin{aligned}
\mathrm{SER}_b &= \frac{1}{N_d}\sum_{t=1}^{N_d}\mathbf{1}\{\hat{\boldsymbol{s}}_t \neq \boldsymbol{s}_t\},\\
\Delta\mathrm{SER} &= \frac{1}{S}\sum_{h=1}^{S}\frac{1}{|\mathcal{W}_h|}\sum_{b\in\mathcal{W}_h}\big(\mathrm{SER}_{h,b} - \mathrm{SER}'_{h,b}\big),\\
\mathrm{RR} &= \frac{\sum_h |\{b\in\mathcal{W}_h : \gamma_{h,b} < \frac{1}{2}\}|}{\sum_h |\mathcal{W}_h|},\\
\mathrm{FRR} &= \frac{\sum_h |\{b\ge 5 : \gamma'_{h,b} < \frac{1}{2}\}|}{\sum_h |\{b\ge 5\}|},
\end{aligned} \tag{11}$$

with $\gamma_{h,b}$ the verdict (25). The paired displacement of level k,

$$D_k(b) = \big\|\theta_b^{(k)} - \theta_b'^{(k)}\big\|, \tag{12}$$

in the Euclidean norm, measures what the attack changed in level k ($k=1$ is the fast level), whether it pushed the level or held it back. Wilson intervals [35] are given for RR and FRR at the headline cell, white-box A1 at SIR 0 dB.

III. CITADEL-RX

A fast level fits every block, and slow levels change only through what the gates accept (Fig. 1); Fig. 2 gives the processing of one block.

A. Levels and the Fast Learner

The weights $\theta \in \mathbb{R}^d$ ($d = 67\,632$ for 4×4 QPSK) are kept at $K=3$ levels with periods $C = (1, 8, 32)$ blocks, all starting at the pre-trained θ^{pre}. With $\theta_b^{(k)}$ the state of level k after block b, the anchor and the surprise-driven step size are

$$a_b = \tfrac{1}{2}\,\theta_{b-1}^{(2)} + \tfrac{1}{2}\,\theta_{b-1}^{(3)}, \tag{13}$$

$$\begin{aligned}
&\Gamma_b = \big\|\nabla_\theta \ell^{\mathrm{det}}(\theta_{b-1}^{(1)}; \tilde{\boldsymbol{Y}}_b)\big\|^2, \quad \bar{\Gamma}_b = 0.9\,\bar{\Gamma}_{b-1} + 0.1\,\Gamma_b,\\
&r_b = \mathrm{clip}\big(\Gamma_b/\bar{\Gamma}_{b-1}, \tfrac{1}{8}, 8\big), \quad \eta_b = \eta_{\max}\frac{2r_b}{1+r_b} \le \frac{16}{9}\eta_{\max},
\end{aligned} \tag{14}$$

with $\eta_{\max} = 5\cdot 10^{-3}$ ($r_b = 1$ and $\bar{\Gamma}_b$ the median of $\Gamma_0,\dots,\Gamma_b$ in the first five blocks). From $\theta_{b-1}^{(1)}$, each stage q in turn [3] takes $T = 300$ steps of

$$\boldsymbol{A} \leftarrow \boldsymbol{A} - \eta_b\,\delta\,\boldsymbol{A}\boldsymbol{\phi}\boldsymbol{\phi}^\top, \tag{15}$$

$$\theta_q \leftarrow \theta_q - \eta_b\lambda\,(\theta_q - a_{b,q}), \tag{16}$$

$$\theta_q \leftarrow \mathrm{Adam}_{\eta_b}\big(\theta_q, \nabla_{\theta_q}\ell^{(q)}(\theta^-; \tilde{\boldsymbol{Y}}_b)\big), \tag{17}$$

where the delta-rule decay (15), $\delta = 0.1$, acts on each weight matrix $\boldsymbol{A}$ with $\boldsymbol{\phi}$ the unit-norm mean of its input, the proximal pull (16), $\lambda = 0.5$, draws stage q toward its part of the anchor, θ^- is the iterate before (15)–(16), and the Adam moments restart with every stage. The result is the fast-adapted state $\check{\theta}_b$.

B. Gated, Clipped Consolidation

The buffer $\mathcal{B}^{(k)}$ holds pairs $(\gamma_{b'}^{(k)}, \theta_{b'}^{(k)})$ indexed by the block b' that stored them; after block b, $(\gamma_b, \theta_b^{(1)})$ enters $\mathcal{B}^{(1)}$ unless the block abstained (abstain state, Section III-D). Then, for $k = 1,\dots,K-1$, if $(b+1) \bmod C_{k+1} = 0$ and $\mathcal{B}^{(k)} \neq \emptyset$,

$$\begin{aligned}
&\theta^{(k+1)} \leftarrow \theta^{(k+1)} + \Pi_{\kappa_{k+1}}\big(\beta\,(m - \theta^{(k+1)})\big),\\
&m = \frac{\sum_{b'}\gamma_{b'}^{(k)}\theta_{b'}^{(k)}}{\sum_{b'}\gamma_{b'}^{(k)}}, \qquad \beta = \eta^{(k+1)}\min\Big\{1, \sum_{b'}\gamma_{b'}^{(k)}\Big\},
\end{aligned} \tag{18}$$

with sums over $\mathcal{B}^{(k)}$, $\eta^{(2)} = 1$, $\eta^{(3)} = \frac{1}{2}$ and no update when the weights sum to zero; the buffer is then emptied, and the new state enters $\mathcal{B}^{(k+1)}$ with the mean weight of the emptied buffer. The clip $\kappa = (8.5, 4.5)$ lies at or just above the largest clean increments on the calibration runs (8.2, 4.5). The first version divided by $\max\{1, \sum_{b'}\gamma_{b'}^{(k)}\}$, which pulls the level toward the origin when nearly every block of a period is rejected and breaks the convexity of Lemma 1; every result here uses (18). The optional drift guard sets $\theta^{(K)} \leftarrow 0.95\,\theta^{(K)} + 0.05\,\theta^{\mathrm{pre}}$ after each level-K update.

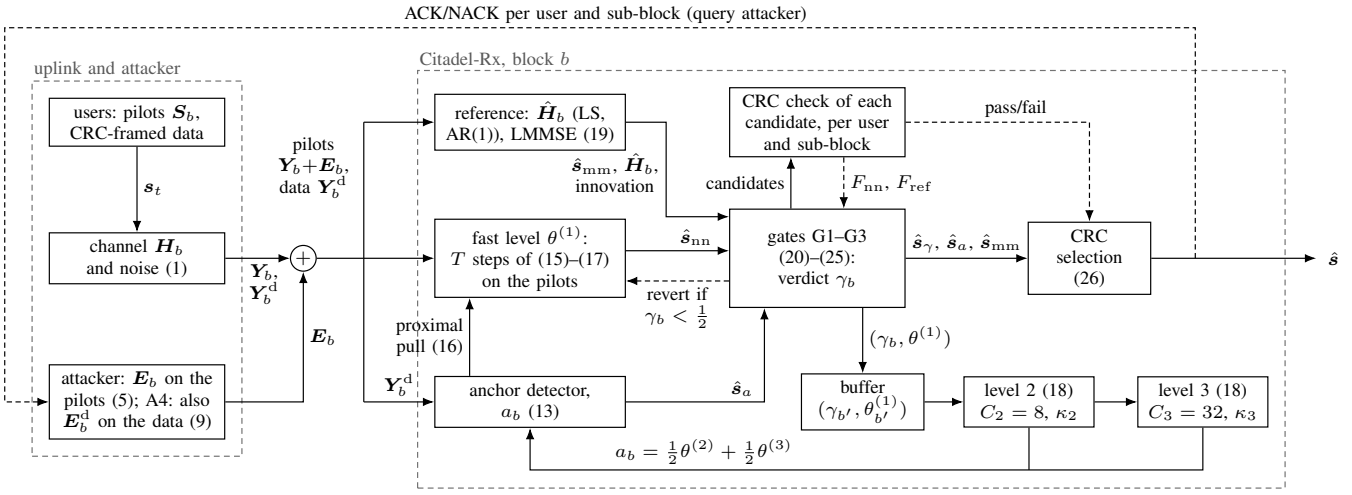


Fig. 1. Citadel-Rx and the uplink with the attacker. Solid arrows pass signals, decisions and states; dashed arrows carry CRC outcomes, the revert command and the ACK/NACK feedback; numbers in parentheses are equations. The anchor detector sees the data of the block but not its pilots. Optional mechanisms are not drawn.

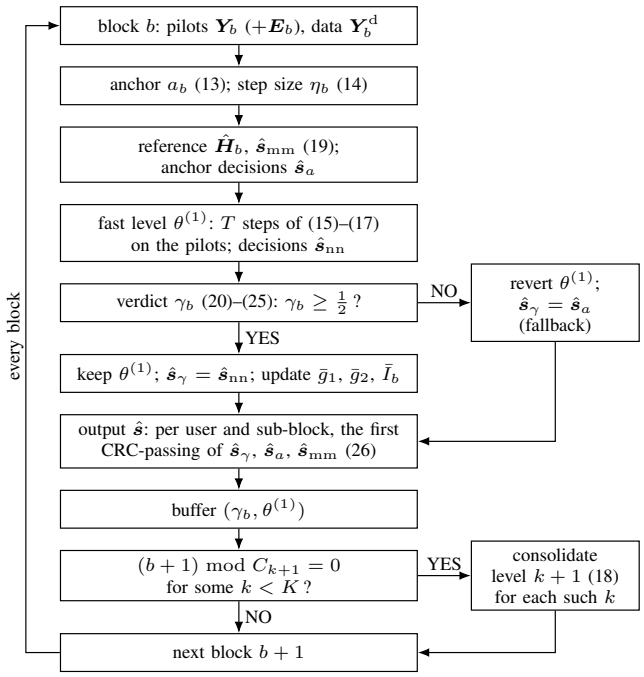


Fig. 2. Processing of one block by Citadel-Rx. Not drawn: $\bar{g}_1$, $\bar{g}_2$, $\bar{I}_b$ are also updated on the first three blocks, an abstained block is rejected and not buffered, and streak recovery may replace a rejected update after five consecutive rejections.

C. Relative Physics Gates

The reference uses no labels. The regularized least-squares (LS) estimate and its fusion with the prediction of a first-order autoregressive (AR(1)) model are

$$\begin{aligned}\hat{\boldsymbol{H}}_b^{\mathrm{LS}} &= \big[(\boldsymbol{S}_b^{\mathrm{H}}\boldsymbol{S}_b + 2\sigma^2\boldsymbol{I}_U)^{-1}\boldsymbol{S}_b^{\mathrm{H}}\tilde{\boldsymbol{Y}}_b\big]^\top,\\ \hat{\boldsymbol{H}}_b &= \varrho\hat{\boldsymbol{H}}_{b-1} + \frac{\mathcal{E}_b^{\mathrm{p}}}{\mathcal{E}_b}\big(\hat{\boldsymbol{H}}_b^{\mathrm{LS}} - \varrho\hat{\boldsymbol{H}}_{b-1}\big),\\ \mathcal{E}_b^{\mathrm{p}} &= (1-\varrho^2)\|\hat{\boldsymbol{H}}_{b-1}\|_F^2, \quad \mathcal{E}_b = \mathcal{E}_b^{\mathrm{p}} + NU\hat{\sigma}_b^2/N_p,\end{aligned} \tag{19}$$

with $\varrho < 1$ the block correlation of Jakes fading ($\varrho = 1$ otherwise, where the gain $\mathcal{E}_b^{\mathrm{p}}/\mathcal{E}_b$ is replaced by one, so that $\hat{\boldsymbol{H}}_b = \hat{\boldsymbol{H}}_b^{\mathrm{LS}}$), $\hat{\sigma}_b^2 = \|\tilde{\boldsymbol{Y}}_b - \boldsymbol{S}_b\hat{\boldsymbol{H}}_b^{\mathrm{LS}\top}\|_F^2/(N_pN)$ the noise power and $\mathcal{E}_b$ the expected innovation energy. LMMSE decisions $\hat{\boldsymbol{s}}_{\mathrm{mm}}$ use $\hat{\boldsymbol{H}}_b$ and $\hat{\sigma}_b^2$; $\hat{\boldsymbol{s}}_a$ and $\hat{\boldsymbol{s}}_{\mathrm{nn}}$ are those of (3) with a_b and $\check{\theta}_b$. The gates compare the model residual, the CRC pass fraction and the innovation:

$$g_{1,b} = \frac{\mathcal{R}_b(\hat{\boldsymbol{s}}_{\mathrm{nn}})}{\mathcal{R}_b^{\mathrm{ref}}\,\bar{g}_{1,b-1}}, \quad \mathcal{R}_b(\hat{\boldsymbol{s}}) = \frac{\sum_t \|\boldsymbol{y}_t - \hat{\boldsymbol{H}}_b\hat{\boldsymbol{s}}_t\|^2}{NN_d}, \tag{20}$$

$$g_{2,b} = \frac{F_{\mathrm{nn}}}{\max\{F_{\mathrm{ref}}, 0.02\}\,\bar{g}_{2,b-1}}, \tag{21}$$

$$g_{3,b} = \frac{I_b}{\max\{\mathcal{E}_b, \bar{I}_{b-1}\}}, \quad I_b = \big\|\hat{\boldsymbol{H}}_b^{\mathrm{LS}} - \varrho\hat{\boldsymbol{H}}_{b-1}\big\|_F^2, \tag{22}$$

with

$$\mathcal{R}_b^{\mathrm{ref}} = \min\{\mathcal{R}_b(\hat{\boldsymbol{s}}_{\mathrm{mm}}), \mathcal{R}_b(\hat{\boldsymbol{s}}_a)\}, \quad F_{\mathrm{ref}} = \max\{F_{\mathrm{mm}}, F_a\}, \tag{23}$$

and $F_X = (JU)^{-1}\sum_{u,j}\mathrm{CRC}_{u,j}(\hat{\boldsymbol{s}}_X)$ the pass fraction of detector X with $\mathrm{CRC}_{u,j} \in \{0,1\}$ the check of sub-block j of user u, $\bar{I}_b$ the running level of I_b on accepted blocks, $\tau_1 = 1.25$, $\tau_3 = \max\{4, \chi^2_{0.999}(NU)/(NU)\} = 4$ here, and τ_2 from (28). G2 follows the attacker's objective, since raising the SER lowers F_{nn}. With LDPC code blocks F_X becomes the virtual pass fraction $(1-\hat{p}_X)^L$, with $\hat{p}_X$ the symbol error rate on the code blocks that decode under a reference. With $\Lambda(x) = 1/(1+e^{-x})$ and $\omega = 0.1$, the verdict is

$$\begin{aligned}&v_{i,b} = \Lambda\Big(\frac{\tau_{i,b} - g_{i,b}}{\omega\,\tau_{i,b}}\Big),\ i \in \{1,3\}, \quad v_{2,b} = \Lambda\Big(\frac{g_{2,b} - \tau_{2,b}}{\omega\,\tau_{2,b}}\Big),\\ &\tau_{i,b} = 1 + (\tau_i - 1)\,\mathrm{clip}(1/r_b, 0.7, 1),\end{aligned} \tag{24}$$

$$\gamma_b = \min\{v_{1,b}, v_{2,b}, v_{3,b}\}, \qquad \bar{g}_{i,b} = \bar{g}_{i,b-1}\,(0.9 + 0.1\,g_{i,b}), \tag{25}$$

so a surprise $r_b > 1$ moves every threshold toward one by up to 30% of its slack. A block is rejected if and only if $\gamma_b < \frac{1}{2}$, and $\bar{g}_{i,b}$, $i \in \{1,2\}$, and $\bar{I}_b$ are updated only on the first three blocks and when $\gamma_b > \frac{1}{2}$.

D. Rejection, Fallback and CRC Selection

A rejected block is reverted and detected with the anchor, which its pilots have not changed; attacked blocks accepted

earlier have entered it through (18). Each sub-block of each user keeps the first of three detections that passes its CRC:

$$(\theta_b^{(1)}, \hat{\boldsymbol{s}}_\gamma) = \begin{cases} (\check{\theta}_b, \hat{\boldsymbol{s}}_{\mathrm{nn}}), & \gamma_b \geq \frac{1}{2}, \\ (\theta_{b-1}^{(1)}, \hat{\boldsymbol{s}}_a), & \gamma_b < \frac{1}{2}, \end{cases} \qquad \hat{\boldsymbol{s}}|_{u,j} = \boldsymbol{c}_{u,j}|_{u,j}, \tag{26}$$

where $\boldsymbol{c}_{u,j}$ is the first of $(\hat{\boldsymbol{s}}_\gamma, \hat{\boldsymbol{s}}_a, \hat{\boldsymbol{s}}_{\mathrm{mm}})$ with $\mathrm{CRC}_{u,j}(\boldsymbol{c}_{u,j}) = 1$, or $\hat{\boldsymbol{s}}_\gamma$ if none passes; all three are already computed for the gates. Optional mechanisms are evaluated with the attacks they answer. Streak recovery re-fits from the anchor on CRC-verified data after five consecutive rejections and keeps the re-fit if it verifies better than the reference. An abstain state, which sets $\gamma_b = 0$ and skips the buffer when the residual on CRC-verified slots leaves its running range, lowered the A4 excess, with overlapping intervals, but abstained after some legitimate redraws (extended version).

E. Design Rule for the CRC Gate

With i.i.d. symbol errors of probability p and a B-bit CRC whose syndrome is uniform on error, a sub-block of L symbols passes with probability

$$\Pr\{\text{pass} \mid p\} = (1-p)^L + 2^{-B}\big(1-(1-p)^L\big). \tag{27}$$

Neglecting 2^{-B} and inverting at F_{ref} gives the rule "reject if $F_{\mathrm{nn}}/F_{\mathrm{ref}} < \tau_2$" with

$$\begin{aligned} p_{\mathrm{ref}} &= 1 - F_{\mathrm{ref}}^{1/L}, \qquad \xi = \max\{2\cdot 10^{-3}, \rho\, p_{\mathrm{ref}}\}, \\ \tau_2 &= \big((1-p_{\mathrm{ref}}-\xi)/(1-p_{\mathrm{ref}})\big)^L, \end{aligned} \tag{28}$$

with $p_{\mathrm{ref}} + \xi$ capped at 0.999. Since $(1-p)^L$ decreases in p, the noiseless ratio falls below τ_2 exactly when $p_{\mathrm{nn}} > p_{\mathrm{ref}} + \xi$. Applied to the normalized ratio (21), ξ counts, to first order, from the fast level's usual error probability, which is the reference's only when $\bar{g}_{2,b-1} = 1$. The evaluated receiver keeps the first-version values $\rho = 1$ and floor $2\cdot 10^{-3}$, which were not derived. The derived slack drops the floor and takes $\tau_2 = \min\{1 - z\sigma_F, \tau_2|_{\xi=\rho p_{\mathrm{ref}}}\}$, with $z = 5$ and σ_F the standard deviation of the normalized ratio on accepted blocks, floored by $\sqrt{(1-F_{\mathrm{ref}})/(J_{\mathrm{c}}F_{\mathrm{ref}})}$ for $J_{\mathrm{c}} = JU$ checks, so that it follows the block length; for independent checks Hoeffding's inequality [36] gives $\Pr\{|\hat{F} - F| \geq x\} \leq 2e^{-2J_{\mathrm{c}}x^2}$ for $x > 0$. The formula (27) is standard; its use is to replace a tuned threshold by a target excess error. As p is pooled over the users, the SER is at most Up, and the independence of errors within a sub-block was not checked against the records.

IV. Retention Bounds

The bounds hold for the algorithm as implemented and a pair of runs on the same channel, noise and seed, with and without an attack from block b_0; the runs coincide before b_0, and primes mark the clean run. Let $n(W) \leq \lceil W/C_K \rceil$ count the blocks $b \in [b_0, b_0 + W)$ with $(b+1) \bmod C_K = 0$.

Lemma 1 (convex consolidation)**.** *If $\sum_{b'} \gamma_{b'}^{(k)} > 0$,* (18) *gives*

$$\begin{aligned} \theta_{\mathrm{new}}^{(k+1)} &= (1-\tilde{\beta})\,\theta^{(k+1)} + \tilde{\beta}\, m, \\ \tilde{\beta} &= \beta \min\big\{1, \kappa_{k+1}/\|\beta(m - \theta^{(k+1)})\|\big\}, \end{aligned} \tag{29}$$

with $\tilde{\beta} \leq \beta \leq \eta^{(k+1)} \leq 1$ and m in the convex hull of the buffered states, so that $\theta_{\mathrm{new}}^{(k+1)}$ is a convex combination of the old state and the buffered states.

Theorem 1 (retention)**.** *(i)* Design bound. *With the clip, for any perturbations and verdicts (with the drift guard, provided both runs consolidate level K at the same blocks, as they do without the abstain state),*

$$D_K(b_0 + W - 1) \leq 2\kappa_K\, n(W). \tag{30}$$

(ii) Structural bound. *Let both runs buffer the same blocks (always so without the abstain state) with $\sum_{b'} \gamma_{b'}^{(K-1)} > 0$ in both runs at every level-K consolidation of the window. Let $\Delta(W)$ be the largest paired displacement of the buffered level-$(K-1)$ states and, at the ith consolidation, $\bar{w}_{i,b'} = \gamma_{b'}^{(K-1)} / \sum_{b''} \gamma_{b''}^{(K-1)}$ and $\bar{w}'_{i,b'}$ the normalized weights, $\tilde{\beta}_i, \tilde{\beta}'_i$ the steps, m'_i the clean gated mean, $R'_i = \max_{b'} \|\theta_{b'}^{\prime(K-1)} - m'_i\|$, and $\theta_i^{\prime(K)}$ the clean level-K state before that consolidation. Then*

$$\begin{aligned} D_K(b_0 + W - 1) \leq \big(\Delta(W) + \Omega(W)\big)\big(1 - (1-\eta^{(K)})^{n(W)}\big) \\ + \sum_{i \leq n(W)} |\tilde{\beta}_i - \tilde{\beta}'_i|\, \big\|\theta_i^{\prime(K)} - m'_i\big\|, \end{aligned} \tag{31}$$

with $\Omega(W) = \max_i \|\bar{w}_i - \bar{w}'_i\|_1 R'_i$. With equal verdicts $\Omega(W) = 0$; if moreover the clip binds in neither run, or with the same factor, at every consolidation of every level, the sum vanishes and $\Delta(W)$ is at most the largest paired fast-level displacement.

Proof. (i) $D_K(b_0 - 1) = 0$, and level K moves only at the $n(W)$ consolidations, by at most κ_K per run; the drift guard multiplies the paired difference by 0.95. (ii) By Lemma 1,

$$\begin{aligned} \theta_{\mathrm{new}}^{(K)} - \theta_{\mathrm{new}}^{\prime(K)} = (1-\tilde{\beta}_i)\big(\theta^{(K)} - \theta^{\prime(K)}\big) + \tilde{\beta}_i\,(m_i - m'_i) \\ + (\tilde{\beta}_i - \tilde{\beta}'_i)\big(m'_i - \theta_i^{\prime(K)}\big). \end{aligned} \tag{32}$$

As both weight vectors sum to one, $m_i - m'_i = \sum_{b'} \bar{w}_{i,b'}(\theta_{b'} - \theta'_{b'}) + \sum_{b'}(\bar{w}_{i,b'} - \bar{w}'_{i,b'})(\theta'_{b'} - m'_i)$ has norm at most $\Delta + \Omega$, so $D \leftarrow (1-\tilde{\beta}_i)D + \tilde{\beta}_i(\Delta + \Omega) + |\tilde{\beta}_i - \tilde{\beta}'_i|\, \|\theta_i^{\prime(K)} - m'_i\|$; iterating from $D = 0$ with $\tilde{\beta}_i \leq \eta^{(K)}$ gives (31), and the drift guard only shrinks the difference. □

Bound (30) holds for every attacker here, including those that pass the gates; its price is that a legitimate change moves the slowest level by at most κ_K per period. The same argument bounds the anchor used by the fallback and the proximal pull,

$$\|a_{b_0+W} - a'_{b_0+W}\| \leq \kappa_2 \lceil W/C_2 \rceil + \kappa_K\, n(W). \tag{33}$$

Bound (31) needs the paired clean run, so it is a diagnostic of an experiment, not a run-time certificate.

Remark 1 (contraction). The regret bound of the first version assumed that the fast block map Φ is L_Φ-Lipschitz in (θ, a) with $L_\Phi < 1$, for the norm $\max\{\|\theta - \theta'\|, \|a - a'\|\}$. With L_θ, L_a its Lipschitz constants in θ and in a, $L_\theta \leq L_\Phi \leq L_\theta + L_a$, and finite-difference probes, which can only underestimate, give $L_\theta + L_a$ from 1.03 to 2.01 on the cosine, Jakes and COST2100 channels and $L_\theta = 1.23$ on the cosine channel; $L_\theta + L_a < 1$ fails, $L_\Phi < 1$ is excluded there, and that bound is withdrawn.

V. Experimental Setup

Scenarios and receivers. Table III defines the scenarios. The receivers are LMMSE and a Huber-robust version [42]; online

TABLE III
SIMULATION SETUP AND PROVENANCE OF THE SETTINGS, READ FROM THE CONFIGURATION FILES AND THE CODE

Item	Value
System	$U \times N = 4 \times 4$, QPSK, SNR 10 dB; 4×16 and 8×32 with the stretched SED profile; 16-QAM at 20 dB (MLP width kept at 256)
Block	$N_p = 100$ pilot and $N_d = 24\,992$ data symbols per user; $N_d = 992$ and 4 992 (nominally 1000 and 5000) and $N_p \in \{32, 100\}$ in Section VI-G
CRC	CRC-8 of [37], $D^8 + D^7 + D^4 + D^3 + D + 1$; $L = 32$; $J = 781$ sub-blocks per user
Channel	SED (2), $P_u = 51, 39, 33, 21$ blocks (users 5–8: 47, 29, 43, 23) [3]; COST2100 traces [38] from the code of [39], in calibration
Run	100 blocks (400, 1000 for A3*; 2000 clean); 40-block window from a block drawn in 10–50 per seed, or fixed: 30–69 (retention), 40–89 (denial), 30–329, 30–929 (A3*)
Seeds	calibration 0–2; test 10–19 (headline cell, final abstain state), else up to 10–14 (stated per table), 10 (sensitivity)
Coded scene	TR 38.901 UMi, 3.5 GHz, 1×4 array, users at 20–150 m, 3 km/h; one block per 1 ms slot; NR LDPC code blocks (base graph 2, $k = 520$, $n = 1000$), CRC-24A [40] (simplified from NR's transport-block CRC), scrambled [41]; 20 belief-propagation iterations from hard symbol decisions
Settings	Q, T, $\eta_{\max}$, width: code of [39]; K, C, $\eta^{(k)}$, λ, δ, τ_i, ω: first version (seeds 0–4); κ: before calibration; slack mode (relative slack), abstain threshold 3, optimizers: calibration; none on test seeds except the abstain statistic (after test runs on seeds 10–12, retested on 15–19) and the stretched SED profile

DeepSIC; Meta-DeepSIC; CRC-Rollback, which is online DeepSIC with G2 at a fixed slack of 0.01 ($\tau_2 = 0.99^L$), judged against LMMSE only, without CRC selection, with rollback to the last accepted weights, and the same with three gates; online DeepSIC with CRC selection; L2-SP [20], elastic weight consolidation (EWC) [21] and an exponential-moving-average (EMA) teacher [22]; CRC-verified decision-directed (DD) training [13], [14]; EATA-style [16] and CoTTA-style [17] receivers; drift detection after [15] with a simpler statistic; a median-of-means gradient after [26]; and a single memory with the fast learner of Citadel-Rx [29]. The joint-symbol backbone of Section VI-G [1], [2] replaces DeepSIC in Citadel-Rx, online training and CRC-Rollback.

Calibration, fixed in advance. Online and Meta-DeepSIC are tuned on clean calibration runs (online: 30 epochs, learning rate 0.001, weight decay 0), and every receiver built on online DeepSIC uses the online setting; the baselines' own method settings are not tuned (L2-SP is swept), and online training on the second backbone is tuned likewise (30 epochs, learning rate 0.001). Of three Citadel-Rx candidates (relative or derived CRC slack, or relative slack with a fast level restarted from the anchor every block and its update projected onto a ball of radius 13), a rule fixed before the runs chose the one with the lowest mean of the clean, A1 and COST2100 A1 window SERs among those with at most 2% clean false rejection, the relative slack. Citadel-Rx keeps its frozen settings on the second backbone and in the coded scene.

Statistics. Intervals are 95% moving-block bootstrap intervals [43] from 4000 resamples of seeds and, within each seed, circular stretches of twice the decorrelation length (a decorrelation length of 6 blocks at the headline cell), at least five blocks long. Tests are exact two-sided Wilcoxon signed-rank tests [44] on per-seed values, whose smallest attainable p-value is 2^{1-S} (0.0625 with five seeds), Holm-adjusted over the four baselines [45]. Wilson intervals treat pooled blocks as independent and are too narrow if consecutive verdicts are correlated. Every attacked run is paired with its clean twin; times (single-threaded runs on an Apple M4 laptop) compare receivers only roughly, so multiply-accumulates are also counted.

TABLE IV
HEADLINE CELL: WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–19. CLEAN: CLEAN-TWIN SER; EXCESS: (11) IN THE WINDOW, 95% MOVING-BLOCK INTERVAL; AFTER: EXCESS IN UP TO 30 BLOCKS AFTER IT; REJ., FALSE REJ.: POISONED AND CLEAN-TWIN BLOCKS REJECTED; Δ CLEAN: CLEAN SER OF CITADEL-RX RELATIVE TO THE BASELINE

Receiver	Clean	Excess [95% CI]	After	Rej.	False rej.	Δ clean
Citadel-Rx	0.0159	$+0.0188$ [0.0142, 0.0246]	$+3.8 \cdot 10^{-4}$	0.993	0.001	–
Online DeepSIC	0.0330	$+0.768$ [0.715, 0.815]	$+0.0189$	–	–	-52%
Meta-DeepSIC	0.0393	$+0.651$ [0.630, 0.670]	$+0.0882$	–	–	-60%
CRC-Rollback	0.0330	$+0.271$ [0.234, 0.307]	$+0.346$	0.897	0.000	-52%
Online + CRC sel.	0.0209	$+0.765$ [0.711, 0.814]	$+0.0118$	–	–	-24%

VI. RESULTS

A. Main Comparison

White-box A1 at SIR 0 dB raises the window SER of Citadel-Rx by $+0.0188$ and that of tuned online DeepSIC by $+0.768$ (Table IV, Fig. 3). The excess of Citadel-Rx is positive on 10 of 10 seeds (exact Wilcoxon $p = 0.002$). The gates reject 397/400 poisoned blocks (Wilson [0.978, 0.997]) and 1/950 clean-twin blocks ([0.000, 0.006]). The excess is also the lowest at SIR ± 10 dB and under transfer (Fig. 3(d)); the closest baseline, CRC-Rollback at $+10$ dB, has $+0.0523$ against $+0.0118$. Seed by seed, Citadel-Rx has a lower window SER and a lower clean SER than each baseline (Holm-corrected $p = 0.008$ for both, the smallest attainable), the clean SER 52% below online DeepSIC, 70% of this gain coming from the CRC selection, which alone does not protect online DeepSIC. Data jamming, a control with 250 times the pilot attack's energy, raises every SER by about the same amount ($+0.813$ to $+0.835$).

B. Failure of Rollback With a Single Memory

CRC-Rollback does not recover after the window at SIR 0 and -10 dB and under transfer ($+0.346$, $+0.637$, $+0.314$). Its reference, LMMSE on the poisoned estimate, passes 0.19 of the CRCs on attacked blocks against 0.71 on the clean twin, so a poisoned fast level that is merely less bad than this reference passes; the first attacked block was accepted on 10 of 10 seeds (seed 10 in Fig. 3(b)) and became the rollback state. Later updates start from it and in most blocks fail the test, so after the window CRC-Rollback rejects a fraction 1.00 of the unattacked blocks (0.000 on the clean twin). Rollback with three gates fails likewise, as does Citadel-Rx with one memory ($K = 1$, $+0.316$). The reference of Citadel-Rx passes 0.76 under attack (0.78 clean) against 0.21 for the poisoned fast level, so only 3/400 attacked blocks were accepted.

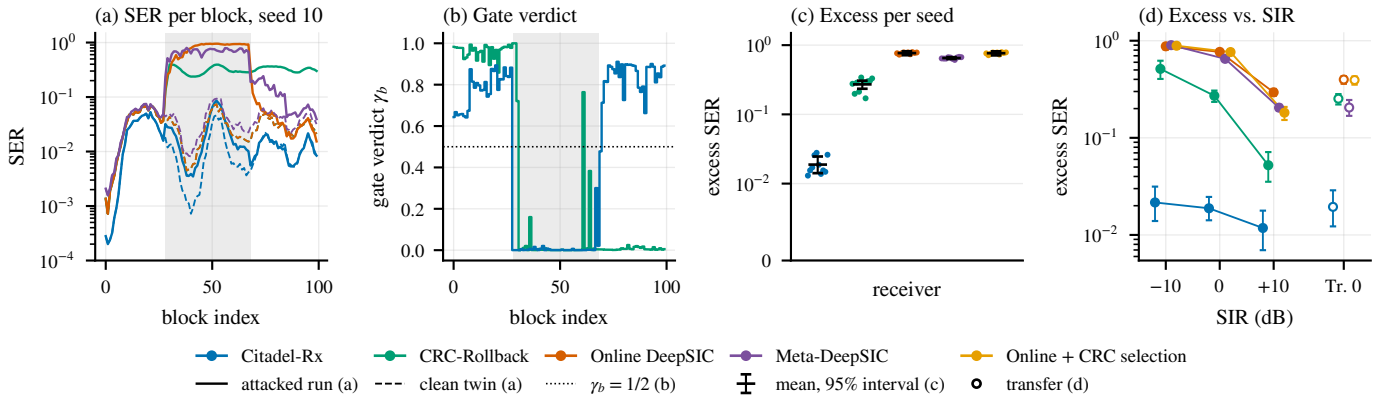


Fig. 3. White-box A1 at SIR 0 dB unless stated. (a) SER per block, seed 10, attacked runs (solid) and clean twins (dashed), window shaded, floored at 10^{-4}; (b) verdict γ_b of CRC-Rollback and Citadel-Rx, seed 10, rejection below $\frac{1}{2}$; (c) excess SER (11) per seed, seeds 10–19, mean and 95% interval (linear below 10^{-2}); (d) excess SER against the SIR (filled) and under transfer at 0 dB (Tr., open), seeds 10–14 except white-box 0 dB, 95% intervals.

TABLE V
ATTACKS BUILT AGAINST THE DEFENSE, SIR 0 DB, TEST SEEDS 10–14, 95% INTERVALS. COMBINED: SOFT GATES, UNROLL 10, PGD 50, 2 RESTARTS, EOT (WITHOUT SOFT GATES FOR ONLINE DEEPSIC). RANK-4 PHYSICAL IS A1 (SAME RUNS AS THE A1 ROW OF TABLE VI). CSI: CHANNEL STATE INFORMATION

Attack	Citadel-Rx excess [95% CI]	Rej.	Online DeepSIC
Unroll 10	+0.0193 [0.0122, 0.0286]	0.995	–
Unroll 50	+0.0192 [0.0120, 0.0284]	1.000	–
PGD 100, 5 restarts	+0.0197 [0.0124, 0.0291]	1.000	–
Soft gates, unroll 10	+0.0193 [0.0121, 0.0288]	1.000	–
Combined	+0.0197 [0.0124, 0.0291]	1.000	+0.858
Rank-1 physical	+0.0171 [0.0114, 0.0241]	0.950	+0.224
Rank-2 physical	+0.0192 [0.0124, 0.0280]	0.995	+0.581
Rank-2, CSI error 0.1	+0.0193 [0.0127, 0.0279]	1.000	+0.518
Rank-2, CSI error 0.01	+0.0195 [0.0127, 0.0282]	0.985	+0.577
Rank-4 physical	+0.0196 [0.0126, 0.0287]	0.995	+0.769
Gray-box	+0.0198 [0.0127, 0.0292]	0.995	–
Query (NACK)	+0.0192 [0.0120, 0.0285]	1.000	+0.522
Signal-referenced budget	+0.0193 [0.0122, 0.0286]	0.990	+0.740

C. Attacks Built Against the Defense

At SIR 0 dB the 13 attacks of Table V give Citadel-Rx +0.0171 to +0.0198, against +0.0196 to +0.0199 for plain A1 in two sets of runs on different machines; the unroll-10, soft-gate and PGD-100 attacks, also run at SIR 10 dB (not tabulated), give +0.0103 to +0.0116. Soft gates (without G3 and the tightening), the query search (not converged) and the other attacks get about as few blocks accepted as plain A1, whereas the rank-1 attacker gets more accepted and gives the lowest excess. With five seeds, differences of a few 10^{-3} lie within the intervals. When nearly every attacked block is rejected, the excess depends on the perturbation's power, not its optimization: it is the price of detecting with an anchor that has not adapted to the block while the LMMSE candidate uses the poisoned estimate. At the headline cell, G2 alone is below one half on 99% of the poisoned blocks, G1 on 88% and G3 on 14%.

D. Attacks on the References and on Adaptation

A4 degrades the references with the fast level (Table VI): at $\alpha = 0.25$ the reference passes 0.65 of the CRCs (0.78 on the clean twin), near enough to the poisoned fast level (0.46) that only 152/200 attacked blocks are rejected. With 20 PGD steps instead of 6, about as many blocks are rejected (0.775 against 0.760) and the excess is higher, with overlapping intervals, so 6 steps may understate this attacker; these rows do not cover every joint attacker, and one with an exact model of the gates was not run. Without attack, a legitimate channel redraw at block 50 makes the gates reject 0.180 of the clean blocks (SER after it 0.250 against 0.224 for online DeepSIC), so the gates slow re-adaptation. The spoof acted as an accepted perturbation during the change, and A1 during the redraw still costs +0.0511 after its window, and +0.0462 with streak recovery (point estimates); online DeepSIC pays +0.168 under the spoof and +0.632 under A1 in the window. The weakest rejected perturbation was not searched for.

TABLE VI
ATTACKS ON THE REFERENCES AND ON ADAPTATION, TEST SEEDS 10–14, WHITE-BOX. A1, A2, A4: SIR 0 DB. SPOOF (SIR 10 DB) AND A1 ON BLOCKS 40–89 WITH A REDRAW AT BLOCK 50: EXCESS OVER THE RUN WITHOUT ATTACK WITH THE SAME REDRAW, NO INTERVAL

Attack	Receiver or state	Excess SER [95% CI]	Rej.
A4, $\alpha = 0.25$	Citadel-Rx	+0.0337 [0.0204, 0.0470]	0.760
A4, $\alpha = 0.25$	20 PGD steps	+0.0478 [0.0279, 0.0689]	0.775
A4, $\alpha = 0.75$	Citadel-Rx	+0.0276 [0.0194, 0.0368]	0.985
A1	Citadel-Rx	+0.0196 [0.0126, 0.0287]	0.995
A2, user 1	Citadel-Rx	+0.0193 [0.0121, 0.0286]	0.995
Spoof, redraw	Citadel-Rx	+0.0968	0.164
A1, redraw	Citadel-Rx	+0.0543	0.988
A1, redraw	+ streak recovery	+0.0461	0.888

E. Retention Under a Slow Small-Budget Attack

Under A1 the largest D_K stays below both bounds in the window (Table VII, Fig. 4(a)), the only stretch the table and the drawn bounds cover; after it D_K rises once more, to 4.85, below the design bound counted to the end of the run (27.0). D_K is the lag of the attacked slowest level behind the clean one: rejected blocks enter with $\gamma_b \approx 0$, so the attacked level moved 4.8 against 8.4, in nearly the same direction (cosine 0.94). A3* at 5% is accepted (1/600 attacked blocks rejected, Table VII); it is the case the bounds are for but not their worst case, and an attacker that maximizes D_K subject to acceptance was not run. After the 300-block A3* attacks the excess is at most +0.0018, and the structural bound is 1.3 to 3.0 times

TABLE VII
RETENTION OF THE SLOWEST LEVEL: LARGEST D_K, STRUCTURAL BOUND (31) AND DESIGN BOUND (30) IN THE WINDOW, MEAN OVER SEEDS; D_K IS BELOW BOTH BOUNDS AT EVERY BLOCK OF THE WINDOW IN EVERY RUN. A1: WINDOW 30–69. A3*: BLOCKS 30–329 OF 400 (900 BLOCKS: 30–929 OF 1000), NORM RELATIVE TO THE PILOTS, TWO SEEDS, SO AN INTERVAL REFLECTS MOSTLY THE VARIATION BETWEEN BLOCKS. GUARD: DRIFT GUARD

Attack	S	Excess SER [95% CI]	Rej.	D_K	Struct.	Design
A1, SIR 0 dB	5	+0.0137 [0.0098, 0.0180]	0.995	4.17	15.7	18.0
A1, SIR 10 dB	5	+0.0073 [0.0048, 0.0099]	0.945	3.73	13.3	18.0
A3*, 5%	2	+0.0017 [0.0014, 0.0021]	0.002	8.48	11.3	90.0
A3*, 20%	2	+0.0122 [0.0099, 0.0147]	0.225	13.30	19.5	90.0
A3*, 50%	2	+0.0109 [0.0086, 0.0136]	0.998	22.03	57.3	90.0
A3*, 50%, guard	2	+0.0110 [0.0087, 0.0137]	0.998	18.72	56.2	90.0
A3*, 5%, guard	2	+0.0017 [0.0014, 0.0020]	0.003	6.86	10.6	90.0
A3*, 5%, 900 blocks	2	+0.0019 [0.0016, 0.0021]	0.001	14.82	16.8	261.0

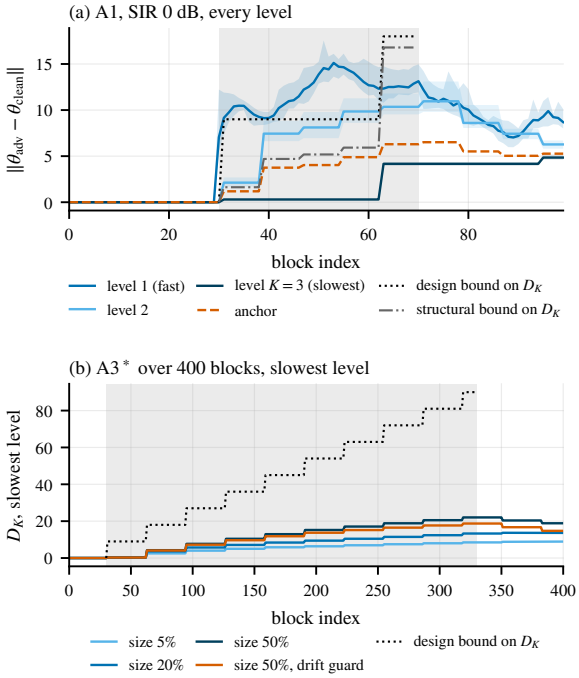


Fig. 4. Paired displacement (12). (a) Levels and anchor under A1 at SIR 0 dB, window shaded, mean over seeds 10–14 with min–max band; design bound (30) and structural bound (31) (largest over seeds) for the slowest level. (b) D_K under A3* (blocks 30–329 shaded) per perturbation size and at 50% with the drift guard, seeds 10–11, with (30).

D_K. The drift guard lowers D_K but not the excess. Over 2000 clean blocks (2 seeds) the SER is 0.0149 over blocks 5–499 and 0.0149 over the last 500.

F. Baselines and Ablations

At their default settings no further baseline resists A1 (excess +0.237 to +0.772, Table VIII). L2-SP buys protection at the cost of adaptation; its sweep stops at weight 1, and its limit, pre-trained DeepSIC without test-time training (zero excess by construction), was not run, so the clean gain of adaptation over it is not measured. The EATA-style filter never fired (entropy at most 0.38 nats against 0.55 nats), so that receiver coincides with EWC. In Table IX, $K = 1$ fails like CRC-Rollback, the fallback is necessary, and G1 or G2 alone suffices. Removing the proximal pull lowers the excess (row "No anchor") only because the clean twin is worse. Without the CRC selection the clean SER doubles while the excess falls to +0.0053, so the selection is most of what the receiver gives up while it rejects. At the headline cell the clip binds in 5/150 clean and 12/150 attacked consolidations, and removing it leaves the excess unchanged; it provides Theorem 1(i), not the measured protection. The reversion, the delta-rule decay and the surprise step are not ablated. On seed 10, scaling any one of eight settings by 0.5, 0.75, 1.5 or 2 keeps the excess between +0.0129 and +0.0146 at a clean false rejection of at most 0.053, above the 2% cap at half the CRC slack and at twice the tightening; on this one seed no setting is critical for the excess within a factor of two.

TABLE VIII
FURTHER BASELINES UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14. CLEAN AND EXCESS SER AS IN TABLE IV; WINDOW SER: ATTACKED RUN IN THE WINDOW

Receiver	Seeds	Clean SER	Excess SER [95% CI]	Window SER
LMMSE	5	0.0562	+0.319 [0.275, 0.362]	0.372
Robust LMMSE (Huber)	5	0.0562	+0.268 [0.226, 0.306]	0.321
Rollback, three gates	5	0.0330	+0.237 [0.194, 0.283]	0.269
L2-SP, weight 0.01	5	0.0380	+0.762 [0.705, 0.814]	0.799
EMA teacher	5	0.0288	+0.438 [0.329, 0.546]	0.466
EWC	5	0.0349	+0.772 [0.720, 0.819]	0.806
CRC-verified DD	5	0.0259	+0.752 [0.671, 0.827]	0.777
CoTTA-style	5	0.0263	+0.245 [0.192, 0.299]	0.271
EATA-style	5	0.0349	+0.772 [0.720, 0.819]	0.806
Drift detection	5	0.0491	+0.686 [0.617, 0.748]	0.735
Single memory (Titans)	5	0.0544	+0.732 [0.703, 0.754]	0.784
Median-of-means gradient	5	0.0300	+0.648 [0.563, 0.726]	0.677
L2-SP, weight 0.001	5	0.0366	+0.785 [0.728, 0.838]	0.821
L2-SP, weight 0.1	5	0.0387	+0.590 [0.551, 0.629]	0.628
L2-SP, weight 1	5	0.0395	+0.136 [0.109, 0.161]	0.175

TABLE IX
ABLATIONS UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14. CLEAN AND EXCESS SER, REJ. AND FALSE REJ. AS IN TABLE IV

Configuration	Seeds	Clean SER	Excess SER [95% CI]	Rej.	False rej.
Complete receiver	5	0.0158	+0.0199 [0.0127, 0.0291]	0.990	0.002
$K = 1$ (gates, rollback, no slow level)	5	0.0340	+0.316 [0.250, 0.396]	0.960	0.086
$K = 2$, $C = (1, 8)$	5	0.0170	+0.0242 [0.0181, 0.0306]	0.985	0.000
No gates	5	0.0158	+0.204 [0.148, 0.265]	0.000	0.000
G1 only	5	0.0158	+0.0218 [0.0145, 0.0309]	0.970	0.000
G2 only	5	0.0157	+0.0194 [0.0124, 0.0284]	0.995	0.000
G3 only	5	0.0158	+0.174 [0.131, 0.218]	0.160	0.000
No anchor ($\lambda = 0$)	5	0.0228	+0.0130 [0.0077, 0.0200]	1.000	0.133
No fallback	5	0.0158	+0.0859 [0.0553, 0.122]	0.995	0.000
No CRC selection	5	0.0317	$+0.0053\ [-1.3 \cdot 10^{-4}, 0.0127]$	0.995	0.000
No consolidation clip	5	0.0158	+0.0194 [0.0123, 0.0285]	0.995	0.000

G. Other Settings, Cost and a Second Backbone

With $N_d = 1000$ the clean false rejection exceeds the 2% cap (Table X), because the floor of the relative slack was set for $J = 781$; the derived slack was not run there. The larger arrays use the stretched SED profile, adopted after test runs with the unstretched one, in which Citadel-Rx also had the lowest excess (+0.0407 at 8×32, against +0.841 for online DeepSIC). Citadel-Rx fails at 16-QAM: its network passes 0.11 of the clean CRCs against 0.93 for LMMSE, so the gates reject 98% of the clean blocks and the clean SER of Citadel-Rx, 0.120, exceeds that of online DeepSIC (0.0512)

TABLE X
OTHER SETTINGS, WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14. ARRAYS: STRETCHED SED PROFILE; 16-QAM: WIDTH 256; JOINT SYMBOL: ONE HIDDEN LAYER OF 512 UNITS, 256 CLASSES. ONLINE: EXCESS OF ONLINE DEEPSIC OR OF ONLINE TRAINING OF THE SAME BACKBONE

Setting	Clean	Citadel-Rx excess [95% CI]	Rej.	False rej.	Online excess
N_d/N_p=1000/32	0.0213	+0.0206 [0.0141, 0.0282]	0.980	0.025	+0.671
N_d/N_p=1000/100	0.0160	+0.0189 [0.0123, 0.0271]	0.995	0.021	+0.763
N_d/N_p=5000/32	0.0202	+0.0215 [0.0142, 0.0310]	0.965	0.006	+0.687
N_d/N_p=5000/100	0.0160	+0.0200 [0.0125, 0.0299]	0.995	0.004	+0.774
$U \times N = 4 \times 16$	$6.8 \cdot 10^{-4}$	+0.0040 [0.0020, 0.0066]	0.910	0.000	+0.818
$U \times N = 8 \times 32$	0.0023	+0.0095 [0.0063, 0.0141]	0.995	0.000	+0.926
16-QAM, 20 dB	0.120	+0.558 [0.478, 0.640]	0.700	0.979	+0.932
Joint symbol	0.0255	+0.0052 [0.0035, 0.0074]	1.000	0.217	+0.349

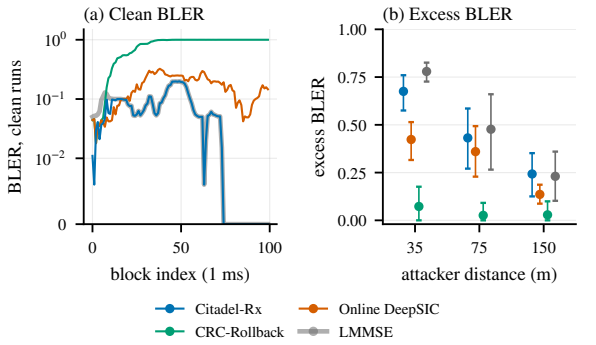


Fig. 5. NR LDPC code blocks over TR 38.901 UMi drops, seeds 10–14. (a) BLER per 1 ms block of the clean twins of the 75 m runs, mean over drops (linear below 10^{-2}); (b) excess BLER in the window against the attacker's distance, 95% intervals; median SIR −26, −3 and +2 dB at 35, 75 and 150 m.

and LMMSE (0.0094); the network is narrower than the 64 hidden units per constellation point used for QPSK, which may contribute, and a wider one was not tested. The full receiver takes 3.03 s per block against 0.56 s for online DeepSIC (4.1 times its multiply-accumulates). On the weaker joint-symbol backbone the ordering holds at 103/475 clean false rejections.

H. A Coded NR Link in a 3GPP Scene

Each test seed is one TR 38.901 urban-micro (UMi) street-canyon drop [46] from Sionna [47] with ideal slow power control; the attacker, one antenna at 35, 75 or 150 m at a user's maximum power, has an SIR set by the geometry, and a stronger one was not run. The defense fails here (Fig. 5). The networks pre-trained on the first block do not follow channel phases that, unlike those of the SED channel, rotate from block to block (clean block error rate, BLER, of online DeepSIC 0.169 against 0.0686 for LMMSE); the phase rotation is the likely cause of this gap, not an established one. The fast level of Citadel-Rx passes a virtual fraction 0.03 against 0.60, so the receiver runs on the anchor and LMMSE; under attack no receiver is protected, and the excess BLER of Citadel-Rx exceeds that of online DeepSIC at every distance, with disjoint intervals at 35 m only; its clean BLER, 0.0650, is the lower of the two. CRC-Rollback has the smallest excess because its clean BLER is already 0.859, and 1.000 over the last 20 blocks, in which LMMSE decodes every code block.

VII. DISCUSSION

What the results support. A test-time receiver can resist pilot poisoning without locating the attacker if what it keeps is checked against something the poisoned pilots of the block do not reach, here the slow memory and the CRC, and if rejected blocks are detected with that memory. The failure of rollback with one memory comes from the reference, not the slack, and the clean gain comes mostly from the CRC selection, which any receiver with a CRC can use.

Where it fails. A4 degrades the references through the data (at $\alpha = 0.25$ the gates reject only 0.760 of its blocks, and it costs +0.0337 against +0.0196 for A1); a network weaker than LMMSE on clean blocks makes the gates reject nearly every clean block; poisoning during a legitimate change denies adaptation; and the gates do not address jamming, which at the same SIR harms every receiver more than poisoning harms Citadel-Rx.

Assumptions. Channel: link level, stochastic UMi drops (not ray-traced) with ideal power control, one narrowband channel per slot and hard-decision LDPC decoding, elsewhere CRC-8 over SED. Seeds: ten at the headline cell, five elsewhere, two for A3* and the long horizon, one for the sensitivity test. Theory: the bounds limit the displacement of the slowest level, not the error rate. Cost: CPU times only, far from real time.

VIII. CONCLUSION

This paper presented Citadel-Rx, a DeepSIC receiver that fits a fast level on every block and changes two slow levels only through physics-gated, clipped consolidation, with an anchor fallback and CRC selection. On held-out seeds, white-box poisoning at SIR 0 dB raised its SER by +0.0188, against +0.768 for online DeepSIC and +0.271 for CRC-Rollback. Adapted attacks left the excess unchanged within its intervals, while A4 (+0.0337; +0.0478 with 20 PGD steps), A1 during a redraw (+0.0543) and the spoof (+0.0968) raised it; A3* at 5%, which the gates accepted (1/600 blocks rejected), cost +0.0017 in SER and left D_K at 8.48, against a clean movement of 26.8 and a design bound of 90.0. Two lessons follow. First, a gate protects a memory only if it judges updates against that memory. Second, relative gates presuppose a network at least as good as the model-based reference on clean blocks. Open directions are references the attacker cannot degrade through the data, adaptation to legitimate change under attack, and networks that track coded-scene channels.

DATA AND CODE AVAILABILITY

Code, configuration files, run records and the scripts that regenerate every number, table and figure are at https://github.com/abraheemrashid99-sys/Citradel-Rx, where `make paper` rebuilds the paper. The extended version, with the complete proofs and every per-scenario table, is in `paper/extended`. The COST2100 traces are those of the public code released with [39].

REFERENCES

[1] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.

[2] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, May 2019.

[3] N. Shlezinger, R. Fu, and Y. C. Eldar, "DeepSIC: Deep soft interference cancellation for multiuser MIMO detection," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1349–1362, Feb. 2021.

[4] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.

[5] K. Wu, W. Jiang, D. Niyato, Y. Li, and C. Luo, "Transfer-based adversarial poisoning attacks for online (MIMO-)deep receviers," arXiv:2409.02430, Sep. 2024.

[6] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 274–283.

[7] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, "On evaluating adversarial robustness," arXiv:1902.06705, 2019.

[8] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1633–1645.

[9] D. Kapetanović, G. Zheng, K.-K. Wong, and B. Ottersten, "Detection of pilot contamination attack using random training and massive MIMO," in *Proc. IEEE 24th Annu. Int. Symp. Pers., Indoor, Mobile Radio Commun. (PIMRC)*, 2013, pp. 13–18.

[10] J. K. Tugnait, "Self-contamination for detection of pilot contamination attack in multiple antenna systems," *IEEE Wireless Commun. Lett.*, vol. 4, no. 5, pp. 525–528, Oct. 2015.

[11] X. Zhou, B. Maham, and A. Hjørungnes, "Pilot contamination for active eavesdropping," *IEEE Trans. Wireless Commun.*, vol. 11, no. 3, pp. 903–907, Mar. 2012.

[12] D. Kapetanović, G. Zheng, and F. Rusek, "Physical layer security for massive MIMO: An overview on passive eavesdropping and active attacks," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 21–27, Jun. 2015.

[13] N. Shlezinger, N. Farsad, Y. C. Eldar, and A. J. Goldsmith, "ViterbiNet: A deep learning based Viterbi algorithm for symbol detection," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3319–3331, May 2020.

[14] S. Schibisch, S. Cammerer, S. Dörner, J. Hoydis, and S. ten Brink, "Online label recovery for deep learning-based communication through error correcting codes," in *Proc. 15th Int. Symp. Wireless Commun. Syst. (ISWCS)*, 2018, pp. 1–5.

[15] N. Uzlaner, T. Raviv, N. Shlezinger, and K. Todros, "Asynchronous online adaptation via modular drift detection for deep receivers," arXiv:2407.09134, Jul. 2024.

[16] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, "Efficient test-time model adaptation without forgetting," in *Proc. 39th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 162, 2022, pp. 16888–16905.

[17] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 7201–7211.

[18] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan, "Towards stable test-time adaptation in dynamic wild world," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[19] L. Yuan, B. Xie, and S. Li, "Robust test-time adaptation in dynamic scenarios," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15922–15932.

[20] X. Li, Y. Grandvalet, and F. Davoine, "Explicit inductive bias for transfer learning with convolutional networks," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 2825–2834.

[21] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[22] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 1195–1204.

[23] T. Wu, F. Jia, X. Qi, J. T. Wang, V. Sehwag, S. Mahloujifar, and P. Mittal, "Uncovering adversarial risks of test-time adaptation," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 202, 2023, pp. 37456–37495.

[24] T. Cong, X. He, Y. Shen, and Y. Zhang, "Test-time poisoning attacks against test-time adaptation models," in *Proc. IEEE Symp. Secur. Privacy (SP)*, 2024, pp. 1306–1324.

[25] Y. Su, Y. Li, N. Liu, K. Jia, X. Yang, C.-S. Foo, and X. Xu, "On the adversarial risk of test time adaptation: An investigation into realistic test-time data poisoning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[26] H. Park, J. Hwang, S. Mun, S. Park, and J. Ok, "MedBN: Robust test-time adaptation against malicious test samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 5997–6007.

[27] Y. Hu and L. Duan, "Theory of continual learning against data poisoning attacks," in *Proc. 43rd Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 306, 2026, pp. 44906–44935.

[28] A. Behrouz, M. Razaviyayn, P. Zhong, and V. Mirrokni, "Nested learning: The illusion of deep learning architectures," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025; arXiv:2512.24695.

[29] A. Behrouz, P. Zhong, and V. Mirrokni, "Titans: Learning to memorize at test time," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025; arXiv:2501.00663.

[30] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 70, 2017, pp. 1126–1135.

[31] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[32] A. Athalye, L. Engstrom, A. Ilyas, and K. Kwok, "Synthesizing robust adversarial examples," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 284–293.

[33] D. Wierstra, T. Schaul, T. Glasmachers, Y. Sun, J. Peters, and J. Schmidhuber, "Natural evolution strategies," *J. Mach. Learn. Res.*, vol. 15, no. 27, pp. 949–980, 2014.

[34] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 2137–2146.

[35] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *J. Amer. Statist. Assoc.*, vol. 22, no. 158, pp. 209–212, Jun. 1927.

[36] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *J. Amer. Statist. Assoc.*, vol. 58, no. 301, pp. 13–30, Mar. 1963.

[37] 3GPP, "Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and channel coding," TS 36.212, V19.1.0, 2025, Sec. 5.1.1.

[38] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin, F. Tufvesson, and P. De Doncker, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.

[39] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Adaptive and flexible model-based AI for deep receivers in dynamic channels," *IEEE Wireless Commun.*, vol. 31, no. 4, pp. 163–169, 2024.

[40] 3GPP, "NR; Multiplexing and channel coding," TS 38.212, V19.2.0, 2025, Secs. 5.1 and 5.3.2.

[41] 3GPP, "NR; Physical channels and modulation," TS 38.211, V19.2.0 (ETSI TS 138 211 V19.2.0, Feb. 2026), Secs. 5.2.1 and 6.3.1.1.

[42] P. J. Huber, "Robust estimation of a location parameter," *Ann. Math. Statist.*, vol. 35, no. 1, pp. 73–101, Mar. 1964.

[43] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Ann. Statist.*, vol. 17, no. 3, pp. 1217–1241, Sep. 1989.

[44] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, Dec. 1945.

[45] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[46] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901, V19.5.0, 2026, Table 7.4.1-1.

[47] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," software, 2022; version 2.2.0, Sep. 2026. [Online]. Available: https://nvlabs.github.io/sionna/